

ANKIT KUMAR

Artificial Intelligence & Machine Learning Engineer

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About Me

Full-Stack AI & ML Engineer with **6+ years experience** specializing in end-to-end design, development and deployment of secure and scalable AI and Machine Learning applications across various cloud environments.

Patent : Method and System for Managing Machine Learning Models Using Dynamic and Parallel Model Operation Platform (Patent Application 202411067665).

Certification : SAFe 6.0 Product Owner Product Manager.

Skills & Technologies

1. Technical Skills

- Languages & Frameworks: Python, C++, SQL, Node.js (Frontend), FastAPI (Backend), FastMCP (MCP)
- GenAI, Machine Learning & Data: LangChain, LangGraph, RAG Pipelines, Multi-Agent Systems, Model Context Protocol, Semantic Search, Embeddings, PyTorch, TensorFlow, Scikit-learn, XGBoost, CNNs, RNNs, PySpark, Pandas, NumPy
- Cloud, MLOps & DevSecOps: AWS, Azure, MLflow, Docker, Terraform (IaC), GitHub Actions, Snyk (SCA/SAST/Container/IaC), SonarQube.

2. Leadership & Product Strategy

- Product Ownership: Certified SAFe 6.0 POPM, Agile, Scrum, Kanban methodologies
- Technical Leadership: Cross-functional team collaboration, stakeholder management.

Professional Experience (6+ Years)

1. MARUTI SUZUKI INDIA LIMITED. | AI/ ML Engineer | Aug 2022 - Present

- *AgentX Platform (DevSecOps Process Automation System)* :
 - Designed DevSecOps process and technical architecture across the MSIL Digital Enterprise and developed an AI-driven (multi-agent) **LangGraph system** to automate complex CI/CD compliance and release pipelines via APIs and MCP. This **accelerated application release cycles by 9%**.
 - To ensure seamless adoption, developed a full-stack application on **AWS** cloud, leveraging a **Node.js** frontend, **FastAPI** backend, **FastMCP** for a **custom MCP server** and integrated Tools APIs including threat modelling, Jira, confluence (CMMI documents), GitHub and Snyk.
- *MLOps Framework [Patented]*:
 - Spearheaded the design and deployment of an end-to-end **MLOps framework** to monitor data quality, drift and overall model health. Successfully onboarded **5+ production projects**, reducing deployment time and **cloud operational expenses by ~20%**.
 - **Filed a patent** : Method and System for Managing Machine Learning Models Using Dynamic and Parallel Model Operation Platform. (Application Number: 202411067665)
- *Finops Platform (Cloud Cost Optimization Framework)* :
 - Designed a standardized **FinOps process and technical architecture** to build a custom multi-cloud (AWS & Azure) web application for cross-divisional cost tracking across the MSIL Digital Enterprise.
 - Engineered an automated FinOps monitoring pipeline to **deliver data-driven cost insights** to executive leadership, **driving a 22% reduction in operational expenses** through proactive EC2 compute rightsizing, Reserved Instance adoption and RDS Graviton migrations.

- *Intelligent Enterprise Knowledge Engine :*
 - Architected an Intelligent Enterprise Knowledge Engine using **LangChain** and **advanced RAG architectures**, reducing complex document retrieval time for senior management and enabling faster data-driven decision-making.
 - Engineered an **automated data ingestion and vectorization** pipeline leveraging FastMCP to standardize context delivery across diverse document formats, ensuring deterministic LLM behavior and improving semantic search fidelity.
- *GenAI Text-to-SQL Dashboard:*
 - Enhanced a natural-language-to-SQL dashboard by integrating **LLM-based query generation** with **schema-aware prompt engineering and retrieval-augmented context**, allowing non-technical business stakeholders to extract data insights directly without writing SQL.
- *Voice Analytics Platform:*
 - Developed an enterprise audio processing engine leveraging **AWS Transcribe** and **OpenAI Whisper** to **analyze 100,000+ (1 Lakh+) daily inbound/outbound voice calls**, delivering real-time KPIs for defect analysis, agent performance, and cross-vertical marketing strategy.
- *Acoustic Brake Diagnostics System:*
 - Built an **RNN-based** system to automate **defect detection for 800+ daily recordings**, saving **30 engineering hours/week** by building model and automating audio **feature engineering** (MFCCs, Spectrograms) and **re-training of model** based on each brake system.
- *Customer Propensity & Clustering:*
 - Increased customer service center retention by training and deploying **unsupervised clustering models** across **13 Pan-India regions**, enabling **2,500+ MSIL dealers** to execute highly personalized vehicle service offers based on historical consumer behavior.

2. ENTYTLE | Data Engineer | June 2020 - Aug 2022

- *Automate Segmentation Service:*
 - Improved and automated the customer segmentation service by implementing the Zipf Law followed by **Ramer-Douglas-Peucker algorithm for complexity reduction**. Utilised python libraries like - rdp, pandas, numpy
- *Dedupe Service:*
 - Improved **unstructured address dataset accuracy by 80%** and reduce manual man-hours, by building a **deduplication data pipeline** from scratch using **PySpark** for preprocessing and using different similarity method to optimize NLP-based name and address matching.
- *Dashboard Visualisation & Data Ingestion:*
 - Streamlined OEM customer data analysis for stakeholders, reducing weekly reporting time, by **automating ETL ingestion pipelines (PySpark, SQL)** and maintaining interactive client dashboards in PowerBI.

Education

- Great Lakes Institute of Management : Post Graduation Program in Data Science & Engineering (2020)
- Goa College of Engineering : Bachelor of Engineering in Electrical & Electronics (2018)